

Birla Institute of Technology and Science, Pilani
 EEE F244 / INSTR F244 Microelectronic Circuits
 Comprehensive Examination, II Semester 2016-2017
Part A: CLOSED BOOK

22

Time: 60 min.

Max. Marks=30

Date: 09-05-2017

R. SANKARA PRASAD.2015A3PS257P.

Unless Given specifically

- Ignore γ , λ in drain current equation.
- Bulk of nmos connected to ground and bulk of pmos connected to V_{dd} . Specify your assumptions.
- All symbols have usual meaning.

Instructions:

- Write the answer to the point in the specified area (mentioned as 'Ans').
- No marks would be given without proper unit of the answer
- No partial marking

- [1] What is the small signal resistance (dV_B/dI_D) offered by n-MOSFET shown in the Fig. 1 at the bias point $V_B=2$ V, $\mu_n C_{ox}(W/L) = 40 \mu A/V^2$ and $V_{THn}=1$ V. Neglect channel length and body bias effect. [Numerical value of dV_B/dI_D is required]. [MM: 2]

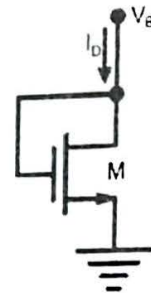


Fig. 1

(Ans): $[dV_B/dI_D] = 25,000 \Omega$

- [2] For the given amplifier circuit in Fig. 2, what is the relationship between output V_1 and V_2 . [MM: 2]

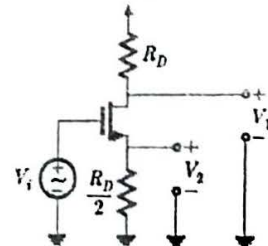


Fig. 2

(Ans): $V_1 = -2V_2$

- [3] Calculate the current (I_D) in a n-channel MOSFET when $V_{GS}=2V_{THn}$ for transistor biasing in saturation region. Consider the following parameters: $V_{THn}=0.75$ V, $W=40 \mu m$, $L=4 \mu m$, $\mu_n=650$ cm²/V-s, $t_{ox}=450 \text{ \AA}$, and $\epsilon_{ox}=3.9 \times 8.85 \times 10^{-14}$ F/cm. [MM: 2]

(Ans): $I_D = 14.02 \mu A$

- [4] Given circuit in Fig. 3, consider, $(W/L)_1=20/0.5$, $I_1=200 \mu A$, $V_{TH0}=0.6 V$, $\Phi_F=0.35 V$, $\mu_n C_{OX}=50 \mu A/V^2$ and $\gamma=0.4 V^{1/2}$. Calculate V_{THn} (considering body bias effect) and V_{out} when $V_{in}=1.2 V$. [Consider M_1 in saturation]. [MM: 2+1]

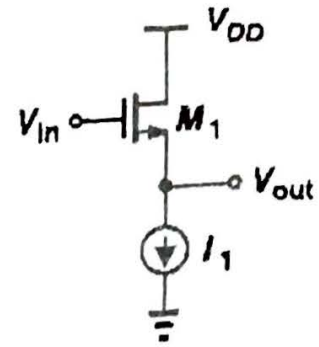


Fig. 3

(Ans): $V_{THn} = 0.542 V$; $V_{out} = 0.2 V$

- [5] For the circuit shown in Fig. 4, MOSFET parameters are $V_{THn}=2 V$, $\mu_n C_{OX}=80 \mu A/V^2$ and $W/L=4$. Find appropriate V_{GS} voltage and choose R_1 and R_2 such that the current in the bias resistors is approximately one-tenth of I_D where $I_D=0.5 mA$ [Assume transistor operates in saturation region]. [MM: 2+1+1=4]

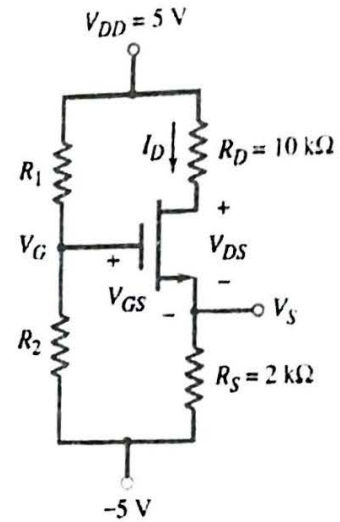


Fig. 4

(Ans): $V_{GS} = 3.76 V$, $R_1 = 104.8 k\Omega$, $R_2 = 95.2 k\Omega$

- [6] Design the circuit in Fig. 5, so that MOSFET operates with $I_D=0.4 mA$ and $V_D=1 V$. The MOSFET parameters are $V_{THn}=2 V$, $\mu_n C_{OX}=20 \mu A/V^2$, $L=10 \mu m$, $W=400 \mu m$ and $\lambda=0$. [MM: 1+2]

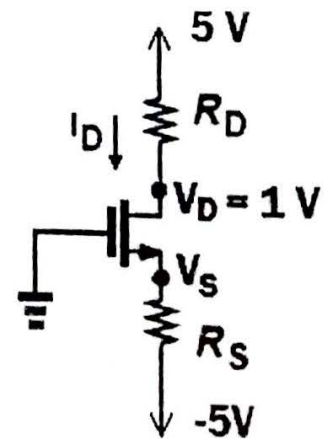


Fig. 5

(Ans): $R_D = 10 k\Omega$, $R_S = 5 k\Omega$

- [7] MOSFET characteristics (Fig.6): $V_{THn} = 1$ V, $\mu_n C_{ox} = 20 \mu A/V^2$, $W/L = 20$ and $\lambda = 0$. Find small signal voltage gain (A_v) of the circuit for $V_{DD} = 5$ V, $R_D = 10$ k Ω and $V_{GS} = 2$ V. Also calculate V_{Dmax} and V_{Dmin} when $v_{gs} = 0.2 \sin(\omega t)$ V. Consider MOSFET works in saturation region. [MM: 2+1+1=4]

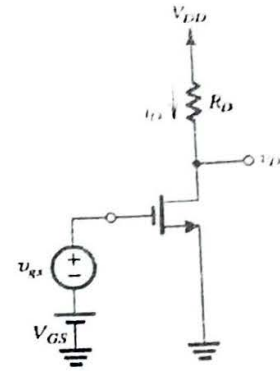


Fig. 6

(Ans): $A_v = 4$ V/V, $V_{Dmax} = 3.8$ V, $V_{Dmin} = 2.2$ V

- [8] Find V_D and R_D so that MOSFET operates with $I_D = 0.4$ mA (Fig.7). Consider: $V_{THn} = 2$ V, $\mu_n C_{ox} = 20 \mu A/V^2$, $L = 10 \mu m$, $W = 100 \mu m$ and $\lambda = 0$. [MM: 2+1]

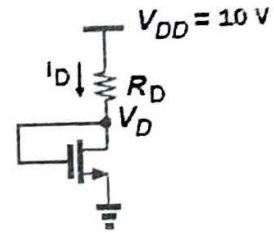


Fig. 7

(Ans): $V_D = 4$ V, $R_D = 1.5$ k Ω

- [9] Given circuit in Fig. 8, $L_1 = L_2 = 10 \mu m$, $W_1 = 100 \mu m$, $W_2 = 300 \mu m$, $\lambda_{np} = 0.1$ V $^{-1}$ and $|V_{TH np}| = 0.4$ V. Calculate V_B for $V_X = 0.9$ V. Consider, $\mu_n C_{ox} = 200 \mu A/V^2$ and $\mu_p C_{ox} = 100 \mu A/V^2$. [MM: 2]

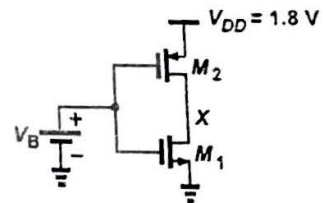


Fig. 8

(Ans): $V_B = 0.95$ V

- [10] Calculate the impedance (in Ohm) looking into node 1 and 2 in the given circuit in Fig.9. [MM: 2]

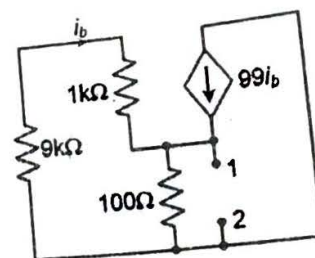


Fig. 9

(Ans): $Z_{1,2} = 50 \Omega$

- [11] Find R_D and drain to source resistance (r_{DS}) of MOSFET for drain voltage $V_D = 0.1$ V (Fig. 10). Consider, $V_{TH0} = 1$ V, $\mu_n C_{ox} W/L = 1$ mA/V² and $\lambda = 0$. [MM: 2+1]

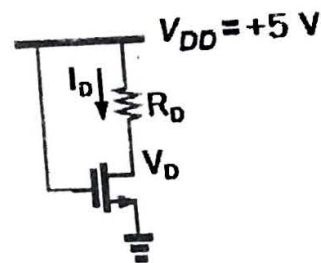


Fig. 10

(Ans): $R_D = 32.667 \text{ k}\Omega$; $r_{DS} = 0.667 \text{ k}\Omega$

*****End*****

Birla Institute of Technology and Science, Pilani
Second Semester 2015-2016
EEE/INSTR F242: Control Systems
Comprehensive Examination (Closed Book)

Date: 13.05.2016

Part B
Time: 2 Hrs

MM: 90

- Q.1 Consider the load position control system given in Fig. Q.1(a). The torque speed characteristic of 2-phase ac servomotor with 115 V applied to control winding is given in Fig. Q.1(b). The various parameters of this system are given below.
- synchro pair sensitivity (K_s) = 100 V/rad
Moment of inertia of the load is 2.5 kg-m² and friction coefficient is 250 Nm/rad/s
Moment of inertia and friction coefficient are negligible.

Gear reduction ratio: $\frac{\theta_L(s)}{\theta_M(s)} = \frac{1}{50}$ and $\frac{\theta_M(s)}{\theta_r(s)} = \frac{1}{1}$

For this System

- Draw the block diagram and
- Calculate the value of amplifier gain K_A such that the system has a steady state error of 0.1 rad for an input velocity of 1 rad/sec. [13]

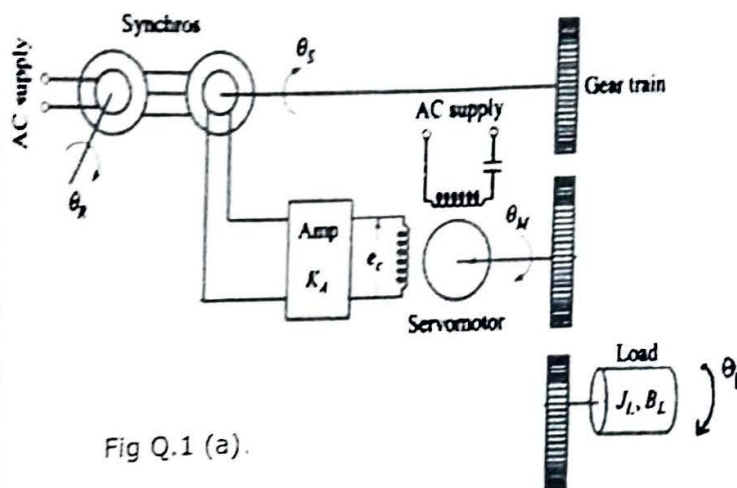


Fig Q.1 (a).

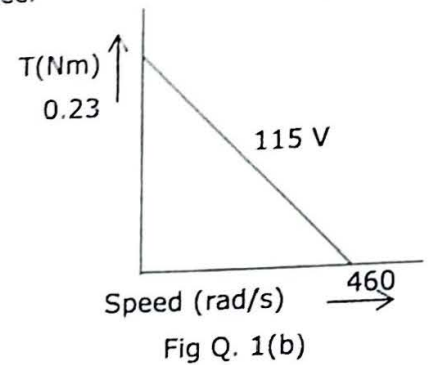


Fig Q. 1(b)

- Q.2 For the liquid level control system shown below,

- Write down the governing differential equations
- Draw the signal flow graph and therefrom determine the transfer function $\frac{\Delta Q_o(s)}{\Delta Q_i(s)}$ and $\frac{\Delta Q_o(s)}{\Delta Q_d(s)}$ using Mason's gain formula. [15]

C_1 and C_2 are the capacities of respective tanks.

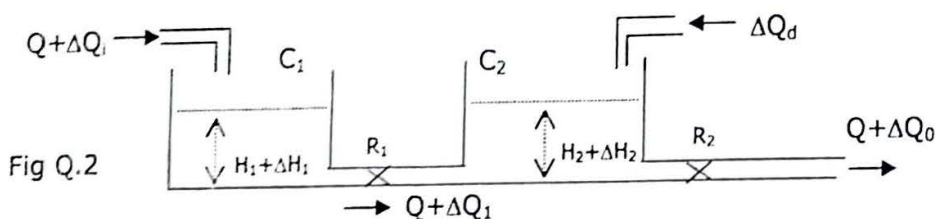


Fig Q.2

PTO

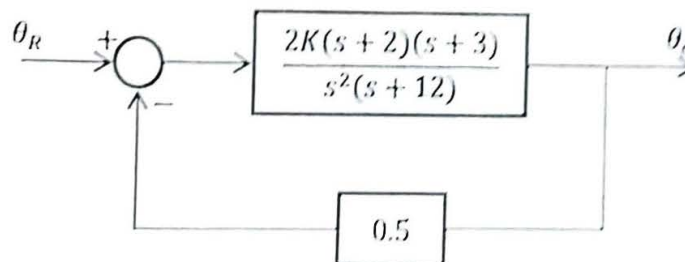
- Q.3 In a unity negative feedback control system, transfer function of the plant is $\frac{50}{s^2 - 20}$. Design a proportional (gain K) and derivative output (gain K_d) controller to meet the following specifications:

Settling time (for 5% tolerance band) ≤ 75 ms and peak overshoot $\leq 25\%$. [10]

- Q.4 Draw the Bode plots for the system whose open loop transfer function is $\frac{K(s+10)}{s(s+2)(s+5)}$ so that the value of steady state error is 10%. From the plots determine gain & phase margin and comment on system stability. Semi-log graph sheet is provided for the same. **Start the plot at $\omega=0.1$ rad/s on log ω axis.** [20]

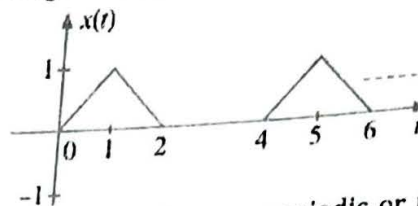
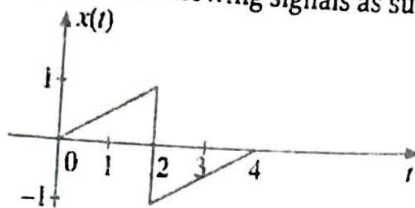
- Q.5 Sketch the Nyquist plot for a system whose open loop transfer function is $\frac{s+0.25}{s^2(s+1)(s+0.5)}$, choosing the appropriate Nyquist contour and therefrom comment on closed loop system stability. [17]

- Q.6 Block diagram of a position control system is shown below. Draw the root locus for this system and determine range of K for which system is under-damped. [15]



- 4 a State and prove the integration property of continuous time Fourier transform. Find Fourier transform of unit step signal $u(t)$. 08
- b Find and sketch the frequency domain response of $x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$ 04
- c Find and sketch the frequency domain response of $x[n] = a^{|n|}$, $|a| < 1$. 06
- 5 a A pressure gauge that can be modeled as an LTI system has a time response to a unit step input given by $(1 - e^{-t} - te^{-t})u(t)$. For a certain input $x(t)$, the output is observed to be $(2 - 3e^{-t} - e^{-3t})u(t)$. For this observed measurement, determine the true pressure input to the gauge as a function of time. (Use Laplace Transform) 06
- b The system function of a causal LTI system is
- $$H(s) = \frac{s+1}{s^2+2s+2}$$
- Determine the response $y(t)$ when the input is
- $$x(t) = e^{-|t|}, \quad -\infty < t < \infty$$
- Plot ROC and comment on stability of system. 06
- c Consider the system S characterized by the differential equation, (use Laplace Transform to)
- $$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t)$$
- a. Determine the zero-state response of this system for the input $x(t) = e^{-4t}u(t)$
- b. Determine the zero-input response of the system for $t > 0^-$, given that
- $$y(0^-) = 1, \quad \left. \frac{dy(t)}{dt} \right|_{t=0^-} = -1, \quad \left. \frac{d^2y(t)}{dt^2} \right|_{t=0^-} = 1.$$
- c. Determine the output S when the input is $x(t) = e^{-4t}u(t)$ and the initial conditions are the same as specified in (b).
- 6 a Given the z-transform pair $x[n] \xleftrightarrow{z} \frac{z}{z^2-16}$, with ROC $|z| < 4$, use the properties to determine the z-transform of the following signals with ROC. 12
- a. $y[n] = x[n-2]$
- b. $y[n] = (1/2)^n x[n]$
- c. $y[n] = x[-n] * x[n]$
- d. $y[n] = nx[n]$
- e. $y[n] = x[n+1] + x[n-1]$
- f. $y[n] = x[n] * x[n-3]$
- b Determine the transfer function and difference equation for impulse response
- $$h[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^{n-2} u[n-1]$$
- Draw the block diagram representation of the system using Direct Form-I and II. 06

- 1 a Express the following signals as sum of singular functions



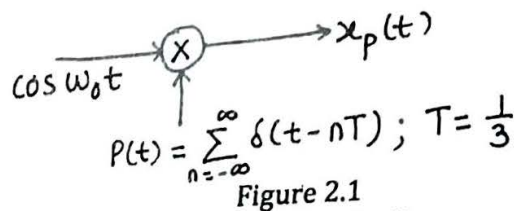
- b Sketch the following signals and determine whether they are periodic or not. Justify your answer. If periodic find the fundamental period.
- $x(t) = 3u(t) + 2 \sin(2t)$
 - $x(t) = 2 + \cos(2\pi t)$
 - $x(t) = \cos(2\pi t + \pi/2) u(t)$
- c Determine whether time-scaling and time-shifting affects the signal energy? Justify your answer in detail.

- 2 a Consider a system in figure 2.1.

- a. Sketch $X_p(j\omega)$ for $-9\pi \leq \omega \leq 9\pi$

For the following values of ω_0

- $\omega_0 = \pi$
- $\omega_0 = 2\pi$
- $\omega_0 = 3\pi$
- $\omega_0 = 5\pi$



- b. For which of the preceding values of ω_0 (in part (a)), is $x_p(t)$ identical?

- b A signal $x(t) = \cos 100\pi t + \cos 250\pi t$ is sampled at sampling frequency f_s and the sampled signal $x_s(t)$ is passed through an ideal low pass filter with bandwidth B . Assuming ideal sampling, sketch the frequency spectrum of the output $y(t)$ of the low pass filter. Also find $y(t)$.

- For $f_s = 150 \text{ Hz}$, and $B = 75 \text{ Hz}$, does aliasing occurs?
- For $f_s = 300 \text{ Hz}$, and $B = 150 \text{ Hz}$, does aliasing occurs?
- If the original signal has to be recovered from the sampled signal, what must be the sampling frequency?

- 3 a Consider a system with input $h(t) = e^t u(t)$ and impulse response $x(t) = e^{2t} u(-t)$ and output $y(t)$.

- Determine $y(t) = x(t) * h(t)$ using graphical method showing sketches of all intermediate steps.
- Verify your result by using suitable transform.

- b If $x(t) = \sqrt{2}(1+j)e^{\frac{j\pi}{4}t} e^{(-1+j/2\pi)t}$. Sketch and label the following:

- $\text{Re}\{x(t)\}$
- $\text{Im}\{x(t)\}$
- $x(t+2) + x^*(t+2)$

M.M:80

Note:

Q1. For an 80286 microprocessor in protected mode of operation, a logical memory address is described by segment selector:offset as 002Dh:0010h. Also, GDTR = 000000h and LDTR = 0010h. Answer following question by referring to the GDT and LDT as given below.

1. Which table will contain the descriptor for this selector?
2. Write the 8 Byte descriptor for this selector, LSB First?
3. What will be the physical address corresponding to this logical address?
4. Which segment (CS/DS/SS) is represented by this descriptor?
5. What will be instruction mode (16/32 bit) for this descriptor?

[2+3+3+2+2=12M]

GDT	00 01 02 03 04 05 06 07
000038h	FE 00 00 01 FF D3 00 00
000030h	FF 01 00 00 20 D2 00 00
000028h	01 00 00 F0 11 E1 00 00
000020h	A0 00 00 AA 01 D2 00 00
000018h	D0 00 01 01 01 D2 00 00
000010h	FF FF 00 01 00 93 00 00
000008h	FF FF 00 00 00 D1 00 00
000000h	A4 01 00 FF FF D2 00 00

LDT	00 01 02 03 04 05 06 07
0140h	FF FF 00 00 00 D1 00 00
0138h	FF 00 00 01 FF D3 00 00
0130h	01 00 00 F0 11 E1 00 00
0128h	FF FF 00 00 4A 1E 0F 00
0120h	FF 00 00 00 80 1E 00 00
0118h	A0 00 00 AA 01 1E 00 00
0110h	A4 01 00 FF FF D2 00 00
0108h	80 01 00 00 FF D2 00 00
0100h	FF FF 00 00 01 D2 0F 00

Q2. There are 8 parking lots in an apartment each can accommodate only one car at a time. Each slot is provided with an optical detector which senses the presence/absence of a car. If the slot is vacant the sensor produces a logic high as the output and a logic low when the slot is occupied. A parking assistance system is to be developed using 8086 and 8255 (only one 8255 is available). The optical sensor signals are bundled into an eight bit bus (signals are compatible with 8086). A display board at the entrance of the apartment (has one indicator for each slot, see fig. below) turns on a GREEN LED for a vacancy in the slot and a RED LED for an occupied slot. Use a base address of 70H.

[8+6= 14M]

(i). Show complete interfacing diagram with all necessary signals.

(ii). Write a well commented code snippet for the above system (use hex-decimal number system)



Q3. The root directory entry is given below:

43 4F 4D 50 52 45 45 58 4E 45 54 16 18 7B BD 4D
4A 3A 4A 3A 00 00 BD BA AB AA FD 01 80 36 00 00

1. Write the size of the file.
2. Write name with extension
3. Write File Attributes
4. The time of file modification
5. The date of file modification
6. The starting cluster number

[2+2+2+2+2+2= 12M]

Q4. If the cluster chain of the file is following

Cluster number	0	1	2	3	4	5	6	7	8	9	A	B	C	D
Next cluster member	FF1	FF6	004	009	008	003	006	FFF	00A	004	00B	FFF	011	FFF

Cluster number	E	F	10	11	12	13
Next cluster member	FFF	FFF	FFF	FFF	FFF	FFF

[3+3+3+3+2= 14M]

(A). What will be the FAT-12 entry for the above cluster chain?

(B). The Boot Sector of a storage device is the following:

EB 3C 90 2A 60 59 60 48 - 49 43 0A 00 40 04 01 00
02 10 01 00 AF F0 09 00 - 0A 00 01 00 00 00 00 00

0110
255 16

(i). What is the size of each sector?

(ii). What is the number of sectors per cluster?

(iii). What is the number of root directory entries?

(iv). What is the number of sectors on disk?

Q5. (i). Write an instruction sequence (ALP snippet) that will cause the priority of an 8259, whose even address is 08H to be IR5, IR6, IR7, IR0, IR1, IR2, IR3, IR4. Solve this problem when the current priority is IR1. Write all the control words in the write sequence. Don't use repetitive rotation commands to change priority and use hex-decimal number system

[8M]

(ii). Now, repeat the above for IR7

(iii). Assuming 16-bit Intel instructions translate (a) from machine to assembly code and (b) translate from assembly code to machine code.

a) 8AF3H

b) MOV [SI], AH

[3+3=6M]

Q6. Using 8254 play "Happy Birthday" music. This has to be achieved by playing the notes D3(147Hz), A3(220Hz), A4(440Hz) for a duration of 250ms, 500ms and 500ms respectively. Given input frequency $F = 1.1931\text{MHz}$, software delay procedure of 250ms and base address is 40H. Now, draw the complete labelled interfacing diagram. Write a well commented code & use hex-decimal number system.

[14M]

Birla Institute of Technology and Science, Pilani
EEE F244 / INSTR F244 Microelectronic Circuits
Comprehensive Examination, II Semester 2015-2016
Part B: OPEN BOOK

Date: 07-05-2016

Time: 140 min.

Max. Marks=80

Unless Given specifically

Take -- $V_{DD} = 3.3V$,

For NMOS device

For PMOS device

$$\mu_n C_{ox} = 140 \mu A/V^2, V_T = 0.7 V, \lambda = 0.1 V^{-1}, C_{ox} = 0.38 pF/\mu m^2$$

$$\mu_p C_{ox} = 40 \mu A/V^2, V_T = -0.8 V, \lambda = 0.1 V^{-1}, C_{ox} = 0.38 pF/\mu m^2$$

NOTE:

If not specified in question,-----

- Ignore γ , λ in drain current equation
- Bulk of nmos connected to ground and bulk of pmos connected to V_{DD} Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFET are biased in saturation region
- All symbols have usual meaning

Q.1

For the MOS amplifier shown in Fig.1, load is PMOS cascode load, all transistors have same $\mu_n C_{ox} W/L$ of $1400 \mu A/V^2$, $V_T = 0.7 V$, dc bias current of $100 \mu A$ and $\lambda = 0.1 V^{-1}$. Assume all transistors are in saturation.

- (a) Find the value of V_{b2} which gives a minimum output voltage. What is the value of minimum output voltage.
- (b) Compute the value of small-signal output resistance (R_{out}). Also find the small-signal low frequency voltage gain (V_{out}/V_{in}).
- (c) Find the expression of small-signal low frequency voltage gain (V_{out}/V_{in}) when from the circuit, transistor M_1 is removed and input voltage (V_{in}) is applied at the source of transistor M_2 .
- (d) Using the circuit of Fig.1, sketch and label the circuit of an NMOS cascode fully differential amplifier. Use current source I_{ss} for current sink and also show the transistor implementation of the given load.

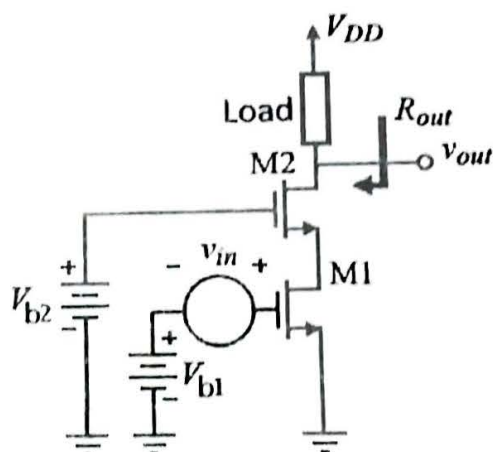


Fig. 1

[MM: 20]

Q.2

- (a) For the circuit shown in the Fig.2(a), all the transistors have same threshold voltage and $L = 1 \mu m$, width $W_1 = 2 \mu m$ and $W_2 = W_3 = 40 \mu m$. $I_{REF} = 25 \mu A$, calculate the current through each transistor. What is the lowest voltage at the output node (V_o) for which current source operation is possible. What are the values of g_m and r_o of Q_2 and Q_3 . Calculate the I_o / I_{REF} with and without considering channel length modulation. **[MM: 10]**

- (b) For the circuit shown in Fig.2(b), identify the circuit of block A, block B and calculated the R_{out} and low frequency voltage gain (V_{out} / V_{in}). All the transistors have same threshold voltage and $L = 1 \mu m$, Width $W_0 = W_1 = 2 \mu m$ and $W_2 = W_3 = W_5 = 10 \mu m$, $I_{REF} = 10 \mu A$. Take $g_{mbs} = 0.1 g_{m5}$. **[MM: 10]**

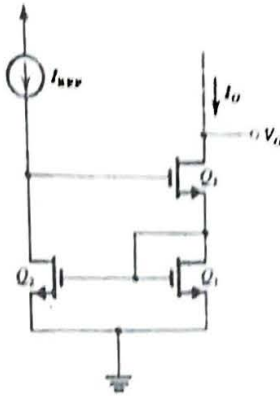


Fig. 2(a)

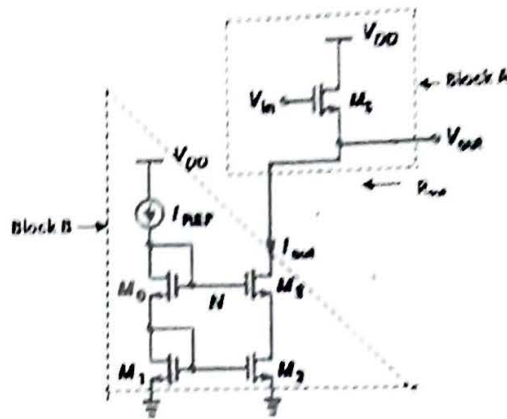


Fig. 2(b)

Q.3 Given Fig. 3, the overdrive voltage of all the transistors is equal $V_{ov}=0.3V$, $|I_{D1,2}| = 30\mu A$. Assume all transistors in saturation. ($\gamma = 0, \lambda \neq 0$)

- Calculate the low frequency voltage gain V_{out}/V_{in} When $R_1 = 0$ and $R_2 = \infty$.
- Calculate the output resistance R_{out} When $R_1 = 0$ and $R_2 = \infty$.
- Identify nature of sampling, mixing and hence the feedback topology ($R_{1,2} = 5k\Omega$).
- Calculate the loop gain ($A\beta$) and hence the closed loop voltage gain and the output resistance using concept of feedback. (Consider $R_{1,2} = 5k\Omega$)

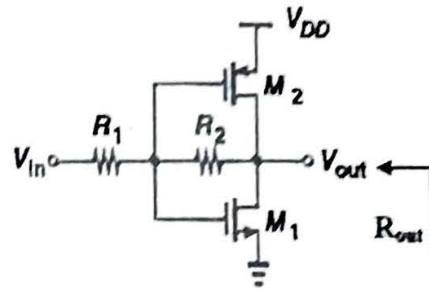


Fig. 3 [MM: 20]

Q.4 Fig.4 shows a two stage CMOS Op-Amp. Assume all the transistors are biased in saturation region.

- Find out the following:
 - dc value of current I_3 , I_5 and I_7
 - Low frequency differential voltage gain (v_o/v_{in}).
 - ICMR (Input common mode range)
 - Output resistance R_o .
 - Gain bandwidth in Hz.
 - Power dissipation
- Derive the expression for RHP (right half plane) zero which occurs in the scheme illustrated in dotted box region of the op amp. Also find out its value.
- Find out the expression for pole at the output node v_o . (Neglect parasitic capacitance).

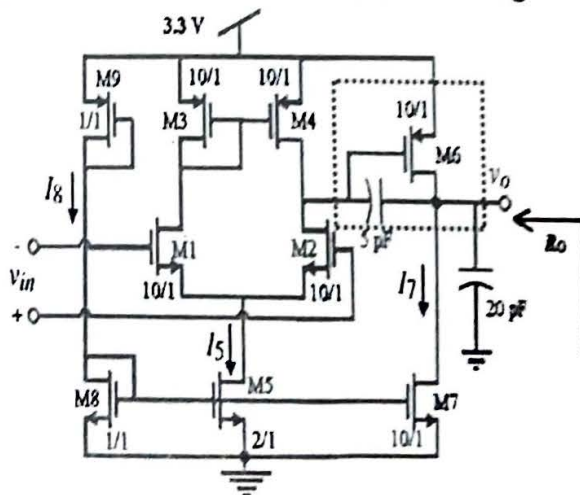


Fig. 4 [MM: 20]

*****End*****